Jihoon Lee

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RESEARCH INTERESTS

I study wireless systems that preserve task performance under communication and computing constraints. My work spans semantic token communication, distributed vehicle control, and GPU sharing for AI-and-RAN systems.

Focus: Semantic token communication; RAN-aware GPU sharing; AI-and-RAN systems; V2X and distributed control; position-aided beam management.

EDUCATION

Ph.D. Program, Electrical & Electronic Engineering

Yonsei University · Mar 2021 - Present

RAMO Laboratory · Advisor: Prof. Seong-Lyun Kim

B.S., Electrical & Electronic Engineering

Yonsei University · Mar 2014 - Feb 2021

Seoul, South Korea

PUBLICATIONS

[1] Hierarchical Token Communication: Semantics-Aware Representation, Protection, and Recovery

Jihoon Lee, Seungeun Oh, Jihong Park, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Transactions on Communications, 2026. Submitted.

[2] Dual-Stage Channel-Posterior Integration for MLM-Aided Token Communications

Hyelin Nam, **Jihoon Lee**, Seung-Woo Ko, Jinho Choi, and Seong-Lyun Kim

IEEE Wireless Communications Letters, 2026. Submitted.

[3] Semantics-Aware Hierarchical Token Communication: Clustering, Bit Mapping, and Power Allocation

Jihoon Lee, Seungeun Oh, Jihong Park, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Global Communications Conference (GLOBECOM), 2026. Accepted for presentation at IEEE GLOBECOM 2026.

PUBLICATIONS (CONTINUED)

[4] Enabling Distributed Control of Vehicle Platooning via Over-the-Air Consensus

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Transactions on Wireless Communications, 2024. Vol. 23, No. 11, pp. 17205-17221.

https://doi.org/10.1109/TWC.2024.3451703

[5] Over-the-Air Consensus for Distributed Vehicle Platooning Control

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE International Conference on Communications (ICC), 2023. Rome, Italy · pp. 5965-5971.

https://doi.org/10.1109/ICC45041.2023.10279484

[6] Stochastic Geometry Analysis of Position aided Beam Management

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim

ICTC Workshop on Post MIMO, 2023. Jeju, South Korea.

Conference paper

IN PREPARATION

RAN-Aware Adaptive GPU Sharing for AI-and-RAN Systems

J. Lee, Y. Jo, Y. Lim, J. Kim, C. Lee, and S.-L. Kim

In preparation · Target: MobiCom 2027

RESEARCH PROJECTS

Development of a 5G-A vRAN Research Platform

Apr 2024 - Present · An integrated research platform that standardizes RU, DU, and CU interfaces and supports software-controlled, virtualized Open RAN systems.

Fundamental Technologies for Semantic Communication

Dec 2024 - Apr 2025 · Research on token-based SLM-LLM interaction to improve communication efficiency and inference accuracy.

Sub-THz Augmented Routing and Transmission for 6G

Dec 2021 - Dec 2023 · Research on routing and transmission foundations for next-generation sub-THz communication systems.

High-Accuracy Positioning-Enabled MIMO for 5G-V2X

Sep 2020 - Dec 2021 · A hardware/software testbed project for position-aided MIMO transmission and networking in 5G vehicle-to-everything services.

DOMESTIC CONFERENCES

English translations of Korean titles.

2025 · A Token-Based Approach to Highly Reliable Semantic Communication

Jihoon Lee et al. · KICS Winter Conference · Oral presentation

2023 · An Over-the-Air Consensus Algorithm for Distributed Vehicle Platooning Control

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim · JCCI · Oral presentation

2021 · Reinforcement-Learning-Based Resource Reallocation Control in Vehicular Networks

Jihoon Lee et al. · KICS Conference Proceedings · Conference paper

2021 · Position-Aware MAC Design for mmWave Vehicular Communications

Jihoon Lee, Seungeun Oh, Jinhyuk Choi, and Seong-Lyun Kim · JCCI · Oral presentation

2020 · Single-Channel Sharing in Massive IoT Using Zigbee

Jihoon Lee et al. · KIEES Summer Conference · Poster presentation

PATENTS

Selected Korean patents. Titles shown in English translation.

Apparatus and Method for Distributed-Consensus-Based Vehicle Platooning Control

KR 10-2023-0009373 · Application · Jihoon Lee and Seong-Lyun Kim

Method and Apparatus for Empty-Bottle Classification and Quantity Estimation Using Multi-Camera Sequential Machine Vision

KR 10-2613465 · Granted · Co-inventor

Method and Apparatus for Position-Information-Based Vehicular Communication

KR 10-2425933 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim

Method and Apparatus for Resource Reallocation Control in Vehicular Networks

KR 10-2433577 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim

TECHNICAL SKILLS & LANGUAGES

Python, MATLAB, System-level simulation, Stochastic geometry, Resource optimization, OAI / NVIDIA Aerial, CUDA Green Contexts

Korean: Native · English: Fluent